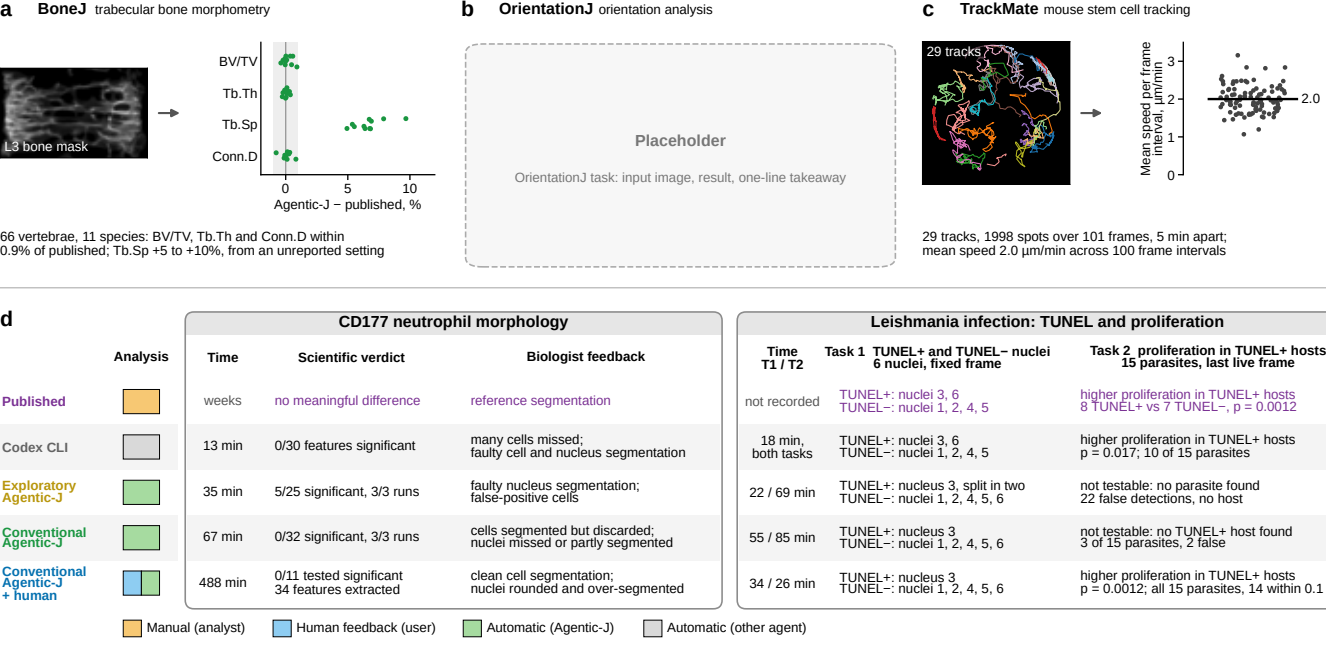


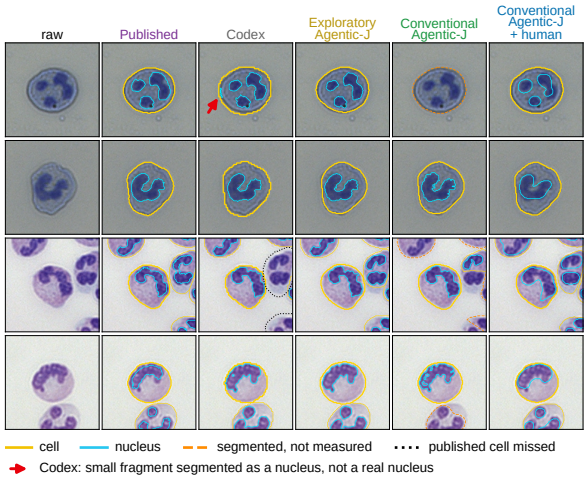
d

Analysis	CD177 neutrophil morphology: Time	Scientific verdict	Biologist feedback
Published	weeks	no meaningful difference	reference segmentation
Codex CLI	13 min	0/30 features significant	many cells missed; faulty cell and nucleus segmentation
Exploratory Agentic-J	35 min	5/25 significant, 3/3 runs	faulty nucleus segmentation; false-positive cells
Conventional Agentic-J	67 min	0/32 significant, 3/3 runs	cells segmented but discarded; nuclei missed or partly segmented
Conventional Agentic-J + human	488 min	0/11 tested significant 34 features extracted	clean cell segmentation; nuclei rounded and over-segmented

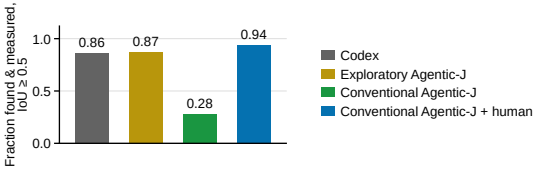
Leishmania infection: TUNEL and proliferation — Time T1 / T2	Task 1 TUNEL+ and TUNEL− nuclei 6 nuclei, fixed frame	Task 2 proliferation in TUNEL+ hosts 15 parasites, last live frame
not recorded	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts 8 TUNEL+ vs 7 TUNEL−, p = 0.0012
18 min, both tasks	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts p = 0.017; 10 of 15 parasites
22 / 69 min	TUNEL+: nucleus 3, split in two TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no parasite found 22 false detections, no host
55 / 85 min	TUNEL+: nucleus 3 TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no TUNEL+ host found 3 of 15 parasites, 2 false
34 / 26 min	TUNEL+: nucleus 3 TUNEL−: nuclei 1, 2, 4, 5, 6	higher proliferation in TUNEL+ hosts p = 0.0012; all 15 parasites, 14 within 0.1

Manual (analyst) · Human feedback (user) · Automatic (Agentic-J) · Automatic (other agent)

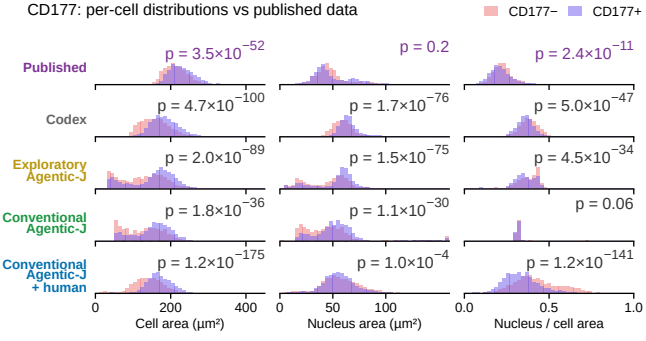
e CD177: segmentations of four example cells



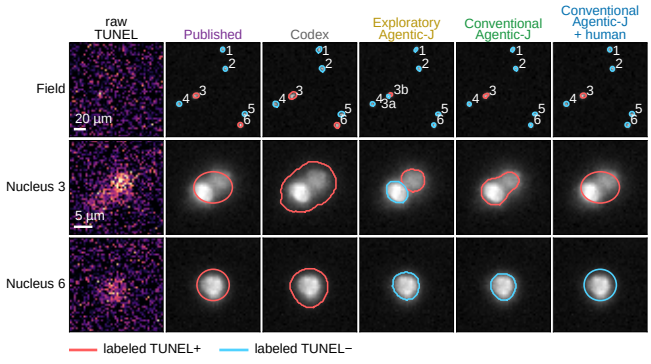
f CD177: published cells found and measured, n = 4389



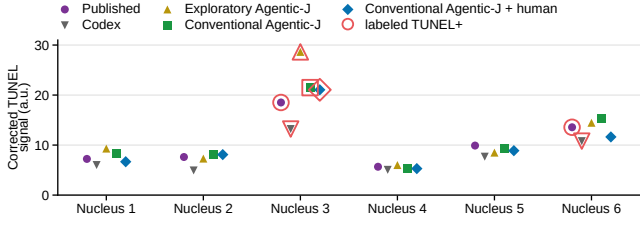
g CD177: per-cell distributions vs published data



h Leishmania task 1: six nuclei, called by each pipeline



i Leishmania task 1: corrected TUNEL signal per nucleus



j Leishmania task 2: proliferation index by host TUNEL status

